Hertie School

LLM in Court: Leveraging Open-Source Language Models to Enhance Empirical Legal Research in Brazil

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"I promise to practice law with dignity and independence, to observe ethics, professional duties and prerogatives and to defend the Constitution, the legal order of the Democratic State, human rights, social justice, the proper application of laws, the rapid administration of justice and ***the improvement of legal culture and institutions****."*

(Oath of the Legal Profession - Art. 20, General Regulation of the Statute of the Lawyer Professorship, Brazilian Bar Association)

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Executive Summary

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Introduction

- context:
  - law is a fundamental aspect in governance and, therefore, to well being
  - however, law is heavely dogmatic and there is little empirical legal research in the country.
  - to do empirical research on law the data mining process is specially complicated, not because the lack of material – Judiciary in BR is full digitalized – but to analyze a large volume of data, with it's specific linguo
  - the first obstacle to a data-driving analysis in law is how to manage text data in large scale and extract structured information from it: one could easily take months to properly analyze few dozens of complex documents.
  - one side-hipothesis: our legal studies and legislative production is suboptimal, due the lack of evidence based decisions
  - relevant to a policy school; in a data science program it's specially relevant to point out solutions to improve this field and offer policy recommendations in the light of the state of the art
- contribution: pipeline, analyse, at least 1 os model trained

There is a significant methodological gap in legal research. Empirical methods are rarely applied, resulting in predominantly qualitative, bibliographical research. When empirical research is done in the legal field, it usually takes a substantial period and a big research team, making it a costly process.

Since the most challenging part of empirical analysis in the legal field is extracting information from very complex documents, I hope to develop a good strategy to make it easier for researchers to analyze how the law is being applied by the courts. Also, I want to apply open-source models since they are a better fit for the public interest and are most easily adoptable by academia and public institutions.

Literature Review

Alves da Silva (2022) paulo: 'O que se entende por pesquisa empírica em direito? A pesquisa empírica em direito é uma pesquisa que toma por objeto não exclusivamente leis e normas, mas principalmente o fenômeno jurídico em sua manifestação concreta' Não olha somente a lei posta, positivada, mas os modos como a lei atua na vida das pessoas e no funcionamento das instituições. A pesquisa empírica em direito, em resumo, analisa o fenômeno jurídico em um sentido mais amplo do que normalmente se lhe atribui na dinâmica de suas manifestações concretas.

Sob tais características, o nosso padrão de produção de conhecimento em direito é predominantemente bibliográfico, baseado em "livros" - em sentido amplo: livros, artigos, opiniões doutrinárias, manuais etc. O jurista é formado para ler e interpretar textos escritos, insculpidos em obras bibliográficas. Desde a faculdade, aprendemos a decifrar uma linguagem rebuscada que veicula um conjunto vasto de conceitos, significados e opiniões e, então, articulá-los em discursos lógicos de função retórica. Se na formação de nível médio aprendemos a decorar e repetir, nas faculdades de direito aprendemos a decifrar e a discorrer

bib review: Cabrera De Bonito et al. (2025) "Embora haja muitas abordagens que tentam lidar com o desafio de transformar textos legais em um conjunto de condições legíveis por máquina, os resultados ainda são insatisfatórios e esse continua sendo um grande desafio em aberto (Dragoni et al., 2016). A pesquisa nas áreas de extração de informações (processamento de linguagem natural, inteligência artificial, entre outros) aumentou o processo de mineração de texto para aprimorar o processo de descoberta de conhecimento neste domínio

(Wagh, 2013)”

“Uma das aplicações de métodos quantitativos ao Direito, em especial, à jurisprudência, é cunhada como jurimetria. O uso dessa ferramenta não é contemporâneo, mas pouco explorada em países de tradição jurídica romano-germânica, como o Brasil. Isso porque háum distanciamento entre pesquisadores de ciências de campos científicos diferentes, aplicando-se aos voltados à análise das ciências jurídicas, cuja abordagem quantitativa dos estudos não é frequente (Andrade, 2018)”

jurimetria Ribeiro et al. (2022) A utilização da estatística aplicada ao direito, embora ainda incipiente no Brasil, já chama a atenção do direito estadunidense há tempos. Para efeito de comparação, o termo jurimetria, quando pesquisado na ferramenta Google Scholar, conhecida por agregar artigos acadêmicos de várias fontes, resulta em 814 resultados (12. set.2021), ao passo que o termo jurimetrics pesquisado na mesma ferramenta retorna 16.600 resultados. Oliver Holmes Jr. (1897) já mencionava que o “homem das leis” do futuro seria aquele que dominasse a estatística e a economia.(“The Path of the Law”, 1897). Desde a década de 1960, o direito americano tem lidado com esse seu aspecto pouco conhecido8 . Entretanto, as possibilidades surgidas com a evolução da informática levaram a jurimetria e a predição probabilística no direito a outro patamar. — jurimetria (18/04): 2 270 — jurimetrics (18/04): 21 800

Está em curso, no Brasil, uma grande revolução tecnológica no judiciário. Os processos passam por uma fase de informatização e digitalização e o Processo Judicial Eletrônico traz, por um lado, desafios aos advogados que precisaram forçosamente renovar e reaprender a peticionar no ambiente digital e, por outro, ao próprio Judiciário, na implementação de sistemas seguros e coesos. Apesar de atualmente todos os processos serem digitais, ainda falta muito para a uniformização dos dados, o que dificulta a pesquisa, principalmente quantitativa, com eles. Um exemplo básico: os números de processos deveriam ser uniformes em todos os ramos da justiça e em todo o Brasil, conforme as determinações da Resolução do CNJ nº 65/2008. Porém, ainda se encontram processos em curso que não seguem essa regra, dificultando o rastreamento de processos

- literature: lawma, tucano, marina lacerda, rayssa, abj drogas Dominguez-Olmedo et al. (2024), Corrêa et al. (2024), Lacerda e Silva (2021), Instituto de Pesquisa Econômica Aplicada (Ipea) & Brasil. Ministério da Justiça e Segurança Pública (2023), Associação Brasileira de Jurimetria (2019), Costa et al. (2011)

The primary focus is the lack of empirical research in the legal field and the costly, time-consuming nature of conducting such studies. While traditional legal research, like that of the Brazilian Applied Economics Institute, and jurimetrics research using data-driven techniques exist, they are limited. In contrast, recent advancements in the application of LLMs to the legal field in the U.S. (e.g., Dominguez-Olmedo et al., Lawma: The Power of Specialization for Legal Tasks) show promise. My thesis will explore the role of open-source models, techniques for their specialization in legal tasks, and the costs of their deployment.

Research Question

At one hand, we recognize a research gap regarding the lack of empirical legal research mainly in Brazil. On the other hand, we see an opportunity to apply the findings of the fast-paced field of natural language processing, empowered by the modern technichs from deep learning. Giving this setting, the research questions relies on: 1. How can large language models be deployed in the Brazilian legal context to accurately extract information, recognize entities, and summarize complex legal documents, considering factors such as cost, computational efficiency, and the specific challenges of legal language and document structure?

This question implies: - is possible to use commercial llms in similar tasks - the new releases on open-source models might reached a point when those are also a perspective

Teorethical Framework

- I'm assuming that the the main contribution would be in the first stage of the data science cicle Wickham et al. (2023)

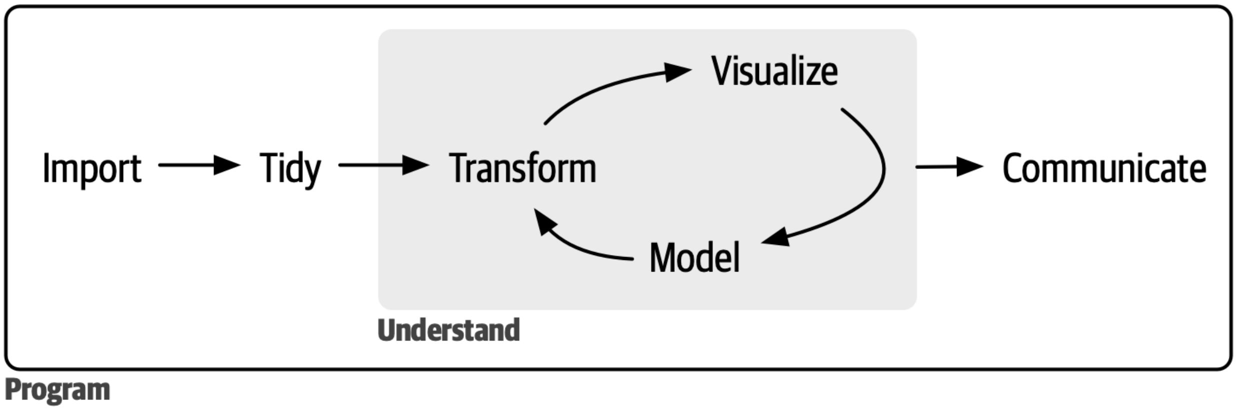

Figure 1: Data science cicle (Wickham et al., 2023)

- NLP: Vajjala et al. (2020): tasks such as "text classification, information extraction, information retrieval, conversational agent, text summarization, question anserwing, machine translation"

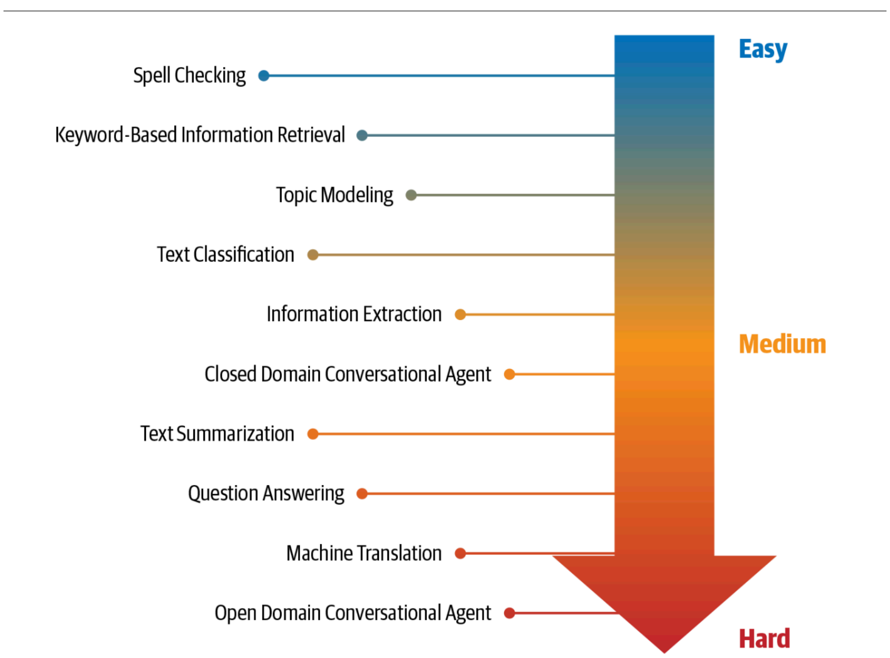

Figure 2: 'NLP tasks organized according to their relative difficulty' (Vajjala et al., 2020)

- LLMs: ‘ushered in a new era for natural language processing (NLP)’; “are deep neural networks models; developed over the past few years”, ‘NLP models performed well in many tasks’ – they focused on specific tasks; ‘failed on follow complex instructions, contextual analysis and creating coherent and contextually appropriate original text’. Raschka (2024)
- chose sota models, both commercial and open source
- gathered data, cleaned and adapted: Lacerda e Silva (2021)
- design a prompt to extract the same features, according with Lacerda e Silva’s dictionary. prompting relies on the ability of the model ‘understand’ specific instructions and act accordingly. So with several iterations, it was possible to guide the model on how to perform the task
- run models
- fine tuned
- test again
- deep dive in the results

* method: - gathered data from judicial sentences in São Paulo - received access from dataset from Lacerda: ± 400 judicial sentences labelled - cleaned data - prompt engeneering - used commercial OS (OpenAI) as baseline - tryied the OS models - llama - gemma - tucano - lawma - lola - libraries HF and unsloth - ft models - compared their performance - limitations: - drug traffic, sao paulo - 2017/2018 - only models that were possible to access - small sample to ft and test - tucano, lawma: - context window was too short to legal docs; - tried with no success to apply rope scaling or yarn - tried RAG to retrieve relevant parts; tried rag to squish the text - possibility: try a task-based prompt – still limited window, but might be sufficient to many documents - lack of computational resources

Data Sources

- data from legal experts – Lacerda e Silva (2021)

Methods

– different tasks, that's why NLP technichs were not feasible – Vajjala et al. (2020): "Recently, large transformers have been used for transfer learning with smaller down-stream tasks. Transfer learning is a technique in AI where the knowledge gained while solving one problem is applied to a different but related problem. With transformers, the idea is to train a very large transformer mode in an unsupervised manner (known as pre-training) to predict a part of a sentence given the rest of the content so that it can encode the high-level nuances of the language in it." –> pick some basemodel and use it to the specific task

– not enought to have an answer, it should be an structured output: JSON. force the output to match a json-style was a challenge, which were overcome with Chase (2022). commercial models are better on this, and do this with the open-source was a challenge

– context window: at least 16k is needed; were insufficient. tryied rope-scaling and yarn but lacked computational resources. Daniel Han & team (2023) was a good and functional option, but it only worked with newer infrastructure.

– Raschka (2024), Lora – Daniel Han & team (2023) enabled using the models

Limitations

– little samples, altough high quality – only used one field of law, only São Paulo – prompt very large – might be some overfitting –

Results

Model catalogue – descriptive metadata

model	family	parameters	type
GPT-4.1	OpenAI	unknown	commercial
GPT-4.1-mini	OpenAI	unknown	commercial
GPT-4o	OpenAI	unknown	commercial
o3-mini	OpenAI	unknown	commercial
Gemma 4.0	Gemma	27 B	open-source
Gemma 4.0	Gemma	12 B	open-source
Gemma 4.0	Gemma	1 B	open-source
Llama 3.1	Llama	8 B	open-source
Llama 3.2	Llama	3 B	open-source
Phi 4.0	Phi	14 B	open-source
Lawma 8B	Lawma	8 B	open-source
Tucano 2.4B	Tucano	2.4 B	open-source

Figure 3: fig1

bla bla

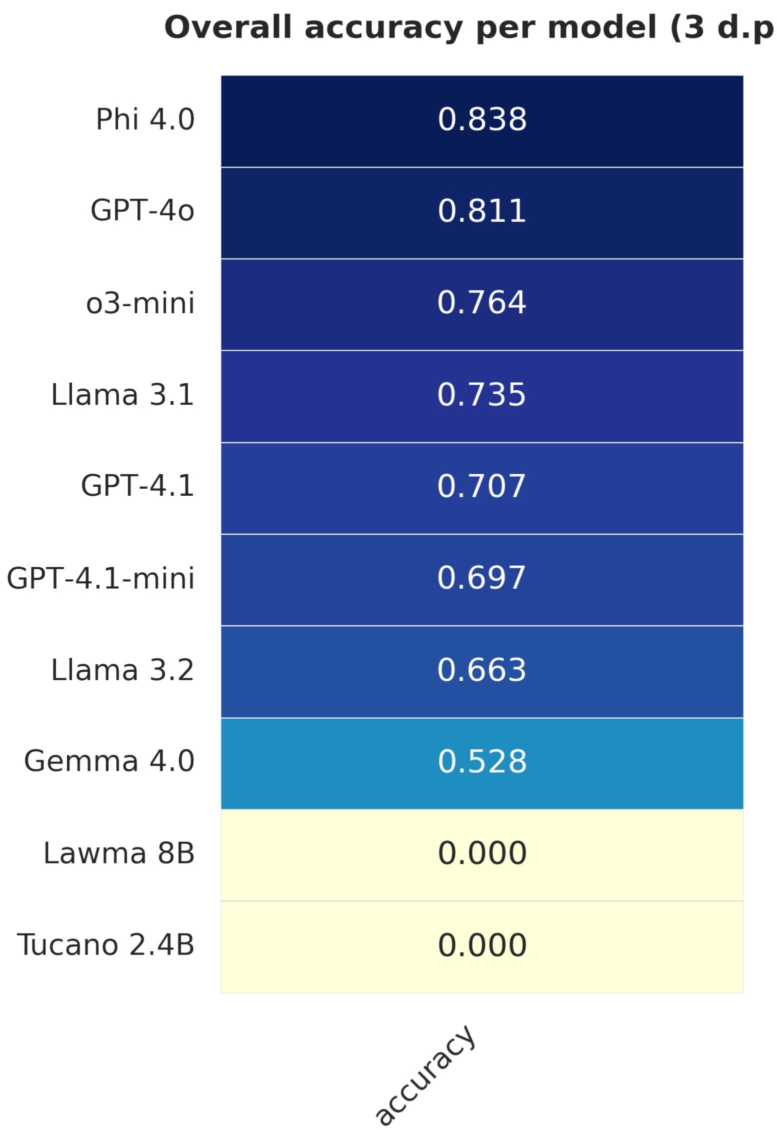

Figure 4: fig2

bla bla

Fine-tuning impact (absolute & relative gains)

model	type	base_acc	ft_acc	Δ abs	Δ %
Llama 3.2	open-source	0.5422804146208	0.78396072013093	0.242	44.6
Phi 4.0	open-source	0.747408619749	0.92798690671031	0.181	24.2
GPT-4o	commercial	0.787779596290	0.93453355155482	0.147	18.7

Figure 5: fig3

Discussion

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Key Findings

- results - os: - cons: difficult to manage, several have short context window; computationaly intensive; - pro: privacy; costs; fast paced; adaptability; full control; - comercial: - cons: privacy; costs; - pro: reliability; easiness; credibility - indeed, specialization is a good strategy - have good potential, altough more testing is needed

Future Research

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- future research:
  - quantization; prunning; distillation;
  - task-based approach
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Policy Recommendations

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- policy recs:
  - academia: open the data from the empirical studies -\-\-> corpus of legal label
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  - os community: larger context windows, more light models

  - judiciary branch: improve metadata recollection and open access to judicial decisoms
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Conclusion

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* conclusions:

  - llms are good tools to leverage empirical legal research in BR

  - commercial models: still up to date a good strategy, despite of its costs and privac

    - pre-process to anonymize is an option

  - os models: feasible and decent baseline performances in certain models; ft is a good
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Alva-Manchego et al. (2021)

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Appendix

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List of experiments